

■ Integrated Secure Data Erasure & Advanced File Recovery Tool

SIH 2026 — Digital Forensics & Data Sanitization

An integrated digital forensics prototype designed to securely sanitize data, recover deleted files, verify san

■ Problem Statement

Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for

The objective of this project is to build a unified platform that combines:

- Secure data sanitization
- File and folder deletion
- Disk-image sanitization
- Filesystem analysis
- Deleted-file recovery
- Raw file carving
- Fragmented-file reconstruction
- Recovery validation
- Recovery confidence scoring
- Recoverability scoring
- Before/after sanitization comparison
- Sanitization verification
- Tamper-evident audit logging
- Automated forensic reporting
- Sanitization certificate generation

The prototype is designed as a focused ****2–3 day MVP**** using controlled test disk images.

■ Core Project Objective

The main idea of the project is:

> ****Don't just erase data. Prove what was recoverable before sanitization — and prove what remains after**

Instead of simply performing data deletion, the system creates a closed forensic loop:

```
```text
RECOVER
↓
MEASURE
↓
BASELINE
↓
SANITIZE
↓
RECOVER AGAIN
↓
COMPARE
↓
VERIFY
↓
REPORT
```

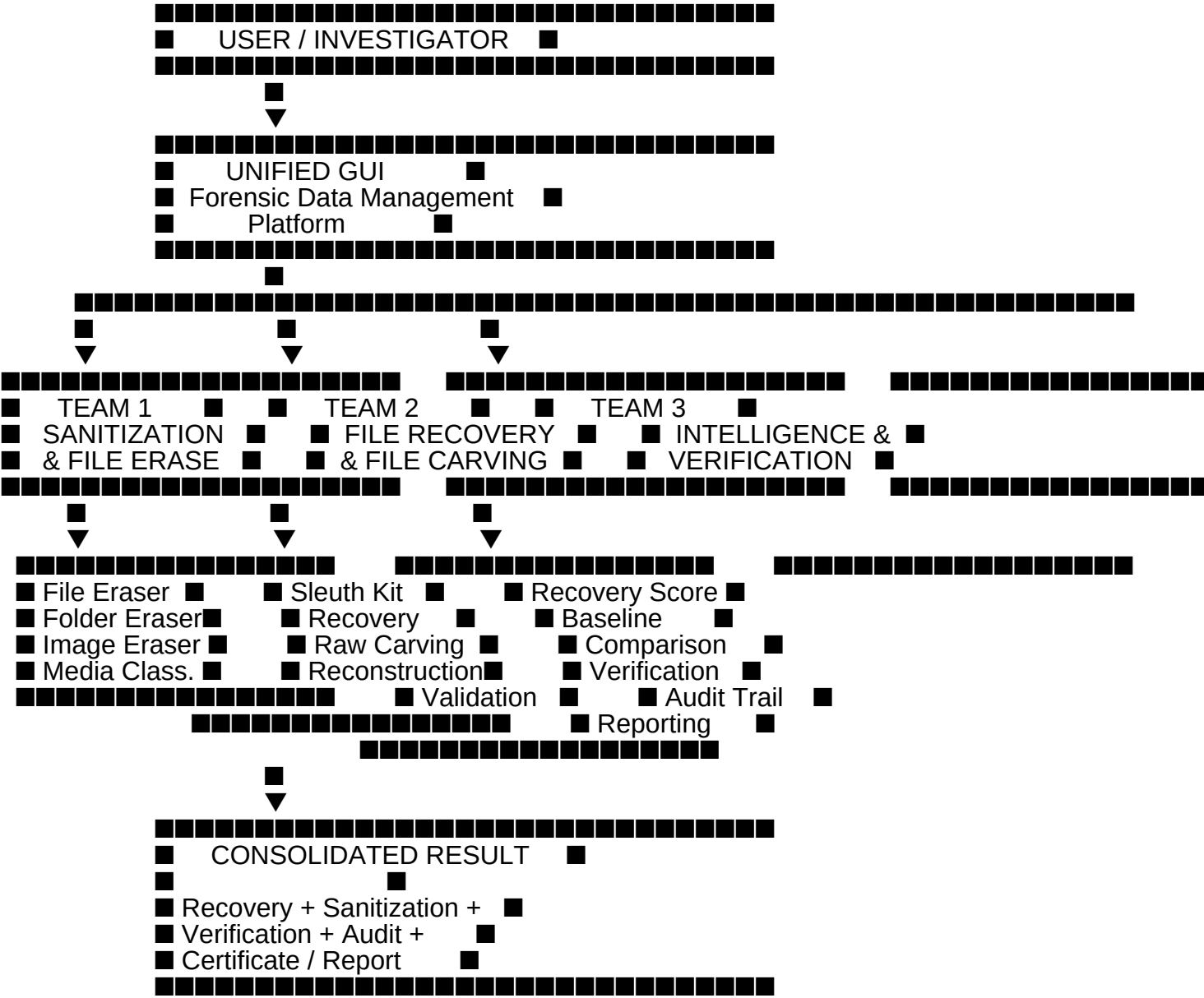
...

This makes the system capable of demonstrating the effectiveness of sanitization using measurable recovery

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# Overall System Architecture

```text



...

Team Structure

The project is divided into three major teams:

| Team | Module | Primary Responsibility |
|--------|---------------------------------------|------------------------|
| Team 1 | Data Sanitization & File Sanitization | Securely remove data |

| Team 2 | File Recovery | Find, recover, reconstruct and validate files |
| Team 3 | Algorithm / Forensic Intelligence & Verification | Score, compare, verify, audit and report |

Each team develops its own module while following common:

- Design concepts
- Visual language
- Data structures
- Integration contracts

The project lead connects all modules into one unified prototype.

■ TEAM 1 — DATA SANITIZATION & FILE SANITIZATION

■ Purpose

Team 1 is responsible for securely removing data and implementing the sanitization layer of the prototype.

The sanitization module is capable of handling:

- Individual files
- Folders
- Controlled disk images

It also classifies the target storage medium and recommends an appropriate sanitization approach.

■ Team 1 Modules

1. File Sanitization

Responsible for sanitizing individual files.

Features

- File sanitization/deletion
- Target identification
- SHA-256 hash capture before sanitization
- Sanitization method selection
- Sanitization status
- Timestamp recording
- Explicit confirmation before destructive operation

2. Folder Sanitization

Responsible for sanitizing folders and their contents.

Features

- Folder selection
- File enumeration
- File sanitization
- Folder-level result reporting

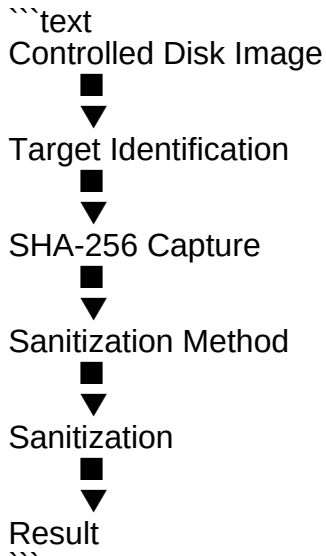
- Target information
- Timestamp
- Sanitization status

3. Controlled Disk-Image Sanitization

Responsible for demonstrating sanitization against controlled forensic disk images.

The MVP should use test disk images rather than automatically sanitizing physical devices.

Example:



4. Storage Media Classification

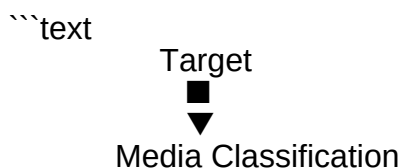
The system identifies the target medium as:

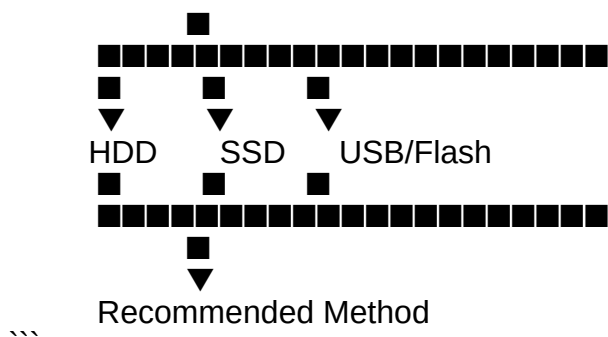
- HDD
- SSD/NVMe
- USB/Flash
- Disk Image
- Unknown

The classifier is used to recommend an appropriate sanitization approach.

5. Sanitization Method Recommendation

The system provides a sanitization recommendation based on the target medium.





The MVP primarily demonstrates classification and recommendation for physical media rather than automa

6. Sanitization Safety Layer

Destructive operations must include safety controls.

Safety requirements

- Explicit user confirmation
- Display target path
- Display SHA-256 before sanitization
- Prevent accidental system-disk targeting
- Controlled test-image operation by default

■ TEAM 2 — FILE RECOVERY

■ Purpose

Team 2 is responsible for finding, reconstructing, recovering, and validating deleted or otherwise recovera

The recovery module combines:

- Sleuth Kit filesystem analysis
- Filesystem metadata analysis
- Deleted-file identification
- Filesystem-based recovery
- Raw file carving
- Fragmented-file reconstruction
- Recovery validation

■ Team 2 Modules

1. Disk Image Analysis

Analyzes the provided forensic disk image.

Responsibilities include:

- Identifying partitions
- Identifying filesystems

- Extracting filesystem information
- Locating relevant evidence

2. Sleuth Kit Integration

The project uses **The Sleuth Kit (TSK)** for filesystem-level forensic analysis.

Important tools include:

```
```text
mmls
fsstat
fls
istat
```
```

General workflow:

```
```text
Disk Image
 █
 ▼
 mmls
 █
 ▼
Partition Information
 █
 ▼
 fsstat
 █
 ▼
Filesystem Information
 █
 ▼
 fls
 █
 ▼
Deleted File Identification
 █
 ▼
Metadata / Inode Analysis
 █
 ▼
Recovery
```
```

3. Deleted File Identification

The system identifies files that are marked as deleted but whose data may still exist in the disk image.

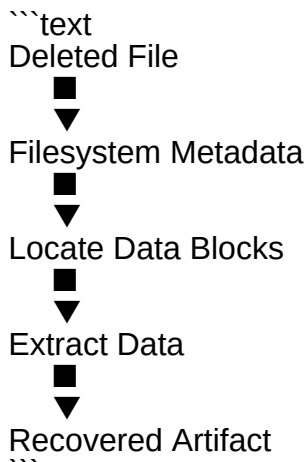
Information may include:

- Filename
- Inode

- File size
- Location
- Allocation status
- Deleted status
- Filesystem evidence

4. Filesystem-Based File Recovery

Deleted files identified through filesystem metadata can be recovered using Sleuth Kit-based methods.



5. Raw File Carving

Raw file carving operates directly on the disk-image bytes.

This approach is useful when filesystem metadata is:

- Missing
- Damaged
- Unavailable
- Insufficient

The carving engine searches for file signatures and extracts file data based on file structure.

■ Initial File-Carving Formats

The initial MVP targets:

| File Type | Extension |
|----------------|-----------------|
| JPEG | `.jpg`, `.jpeg` |
| PNG | `.png` |
| PDF | `.pdf` |
| Microsoft Word | `.docx` |
| ZIP | `.zip` |

■ Raw File Carving Pipeline

```text

Disk Image



Read Raw Bytes



Search File Signatures



Identify File Header



Determine File Structure



Locate Appropriate End Marker



Extract File



Validate File



Generate Metadata

```

6. Fragmented File Reconstruction

A file may be stored in multiple non-contiguous locations on a disk.

Example:

```text

[FILE HEADER]

[FILE DATA]

[RANDOM DATA]

[RANDOM DATA]

[FILE DATA]

[FILE DATA]

[FILE END]

```

The system attempts to reconstruct fragmented files where sufficient structural information is available.

Possible evidence:

- File signatures
- File structure
- Filesystem metadata
- Block/cluster information
- Structural markers

- Validation results

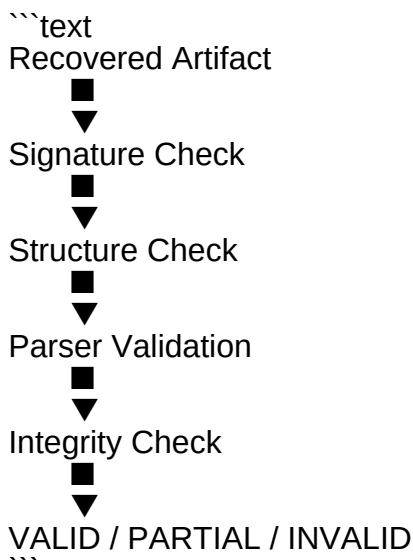
7. Recovered File Validation

A recovered file is not automatically considered valid just because bytes were extracted.

The validation layer checks whether the recovered data represents a usable file.

Possible checks include:

- File signature verification
- Structural validation
- Parser validation
- Successful decoding
- Successful opening
- File size verification
- SHA-256 calculation
- Integrity status



8. Recovery Metadata

Each recovered artifact should contain metadata such as:

```
```json
{
 "file_name": "recovered_001.jpg",
 "file_type": "JPEG",
 "source_image": "carving.img",
 "offset": 1048576,
 "size": 245760,
 "recovery_method": "raw_carving",
 "validation_status": "VALID",
 "sha256": "..."
}
```

This metadata becomes evidence for Team 3.

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## # ■ TEAM 3 — ALGORITHM / FORENSIC INTELLIGENCE & VERIFICATION

### ## ■ Purpose

Team 3 converts recovery and sanitization results into measurable and explainable forensic evidence.

Team 3 is responsible for:

- Recovery confidence
- Recoverability scoring
- Baseline creation
- Before/after comparison
- Sanitization verification
- Audit trail
- Reporting
- Certificate generation

---

## # ■ Team 3 Modules

### ## 1. Explainable Recovery Confidence

The system produces a recovery confidence score supported by understandable evidence.

Possible evidence includes:

- Valid file signature
- Structural checks
- Parser validation
- Filesystem evidence
- Successful file decoding
- Integrity checks

The goal is not just:

```
```text
FILE FOUND
```
```

but:

```
```text
FILE FOUND
  ■
  ■■■ Valid Signature
  ■■■ Valid Structure
  ■■■ Parser Passed
  ■■■ Filesystem Evidence
  ■■■ Integrity Passed
  ■
  ▼
  RECOVERY CONFIDENCE
```
```

---

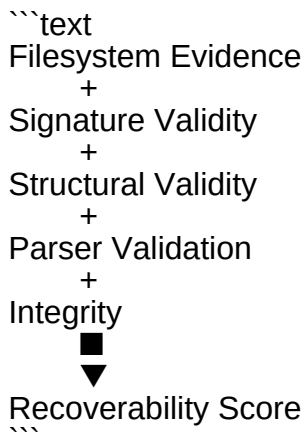
---

## # 2. Recoverability Score

The project defines a **Recoverability Score**.

This score summarizes how recoverable an artifact is rather than simply reporting that the file was found.

Example conceptual model:



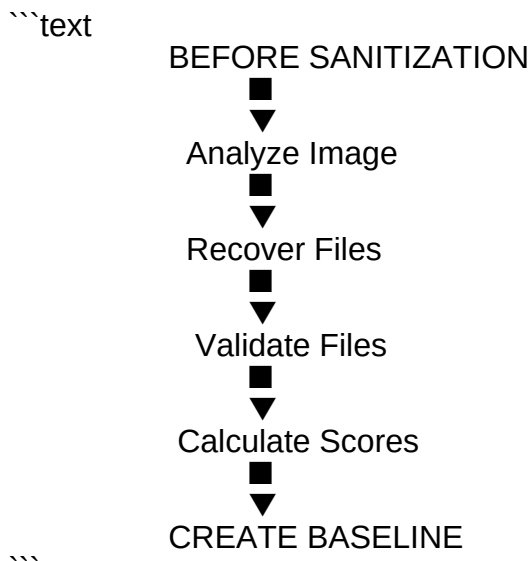
The exact scoring algorithm is project-defined.

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## # 3. Pre-Sanitization Recovery Baseline

Before sanitization, the system performs recovery and records the identified artifacts.

Example:



The baseline records what was recoverable before sanitization.

---

#### # 4. Post-Sanitization Recovery

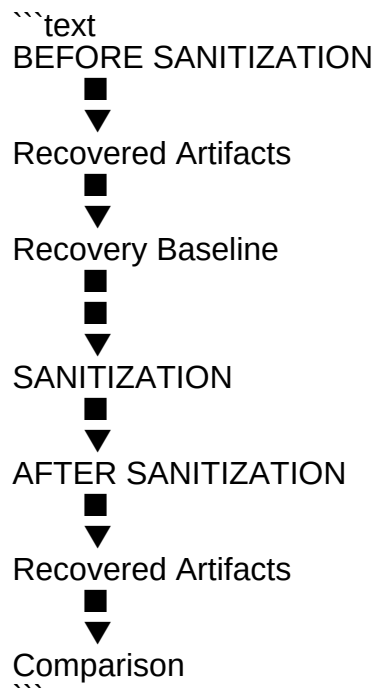
After Team 1 sanitizes the target, Team 2 performs recovery again.



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#### # 5. Before/After Comparison

Team 3 compares:



The system determines whether previously identified artifacts remain recoverable.

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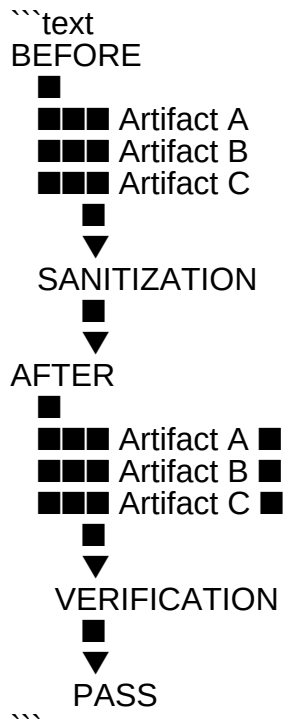
## # 6. Sanitization Verification

The system verifies the sanitization outcome based on the defined recovery procedure.

The verification statement is:

> **"No previously identified artifacts were recovered under the defined verification procedure."**

The system can produce:

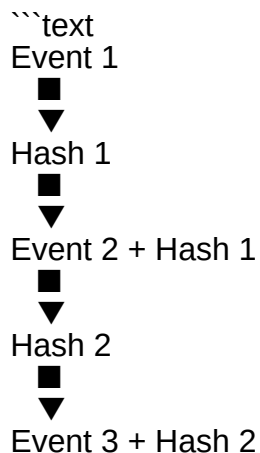


If previously identified artifacts remain recoverable, the verification result can be marked as **\*\*FAIL\*\***.

## # 7. Tamper-Evident Audit Trail

The system maintains a hash-chained audit trail.

Example:





If an earlier recorded event is modified, the hash chain becomes inconsistent.

This provides an integrity mechanism for the recorded workflow.

> This is a tamper-evident audit mechanism and **\*\*not blockchain\*\***.

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## # 8. Automated Forensic Report

The system generates a consolidated report containing:

- Target information
- Recovery findings
- Recovery metadata
- Recovery confidence
- Recoverability score
- Sanitization details
- Before/after results
- Verification outcome
- Audit integrity
- Final status

---

## # 9. Sanitization Certificate

The system can generate a sanitization certificate/report containing the final verification outcome.

Example:

```
```text
=====
      SANITIZATION CERTIFICATE
=====
```

Target : carving.img

Pre-Sanitization
Artifacts Recovered : 5

Sanitization Status : COMPLETED

Post-Sanitization
Artifacts Recovered : 0

Verification : PASS

Audit Integrity : VALID

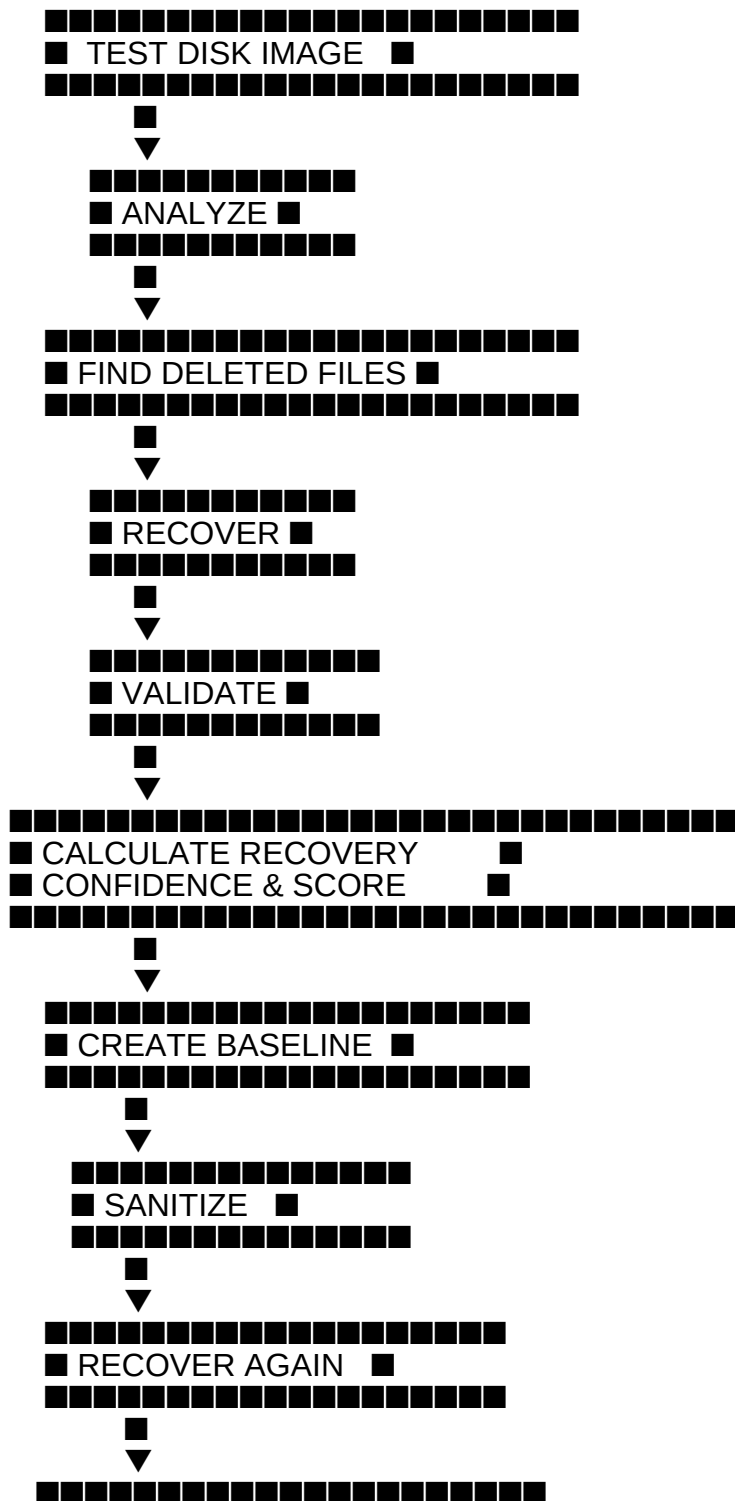
Statement:
No previously identified artifacts were recovered
under the defined verification procedure.

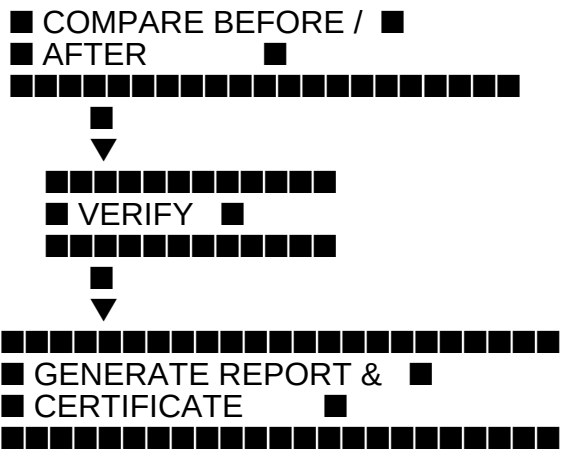
=====

■ COMPLETE SYSTEM WORKFLOW

The complete integrated workflow is:

```text





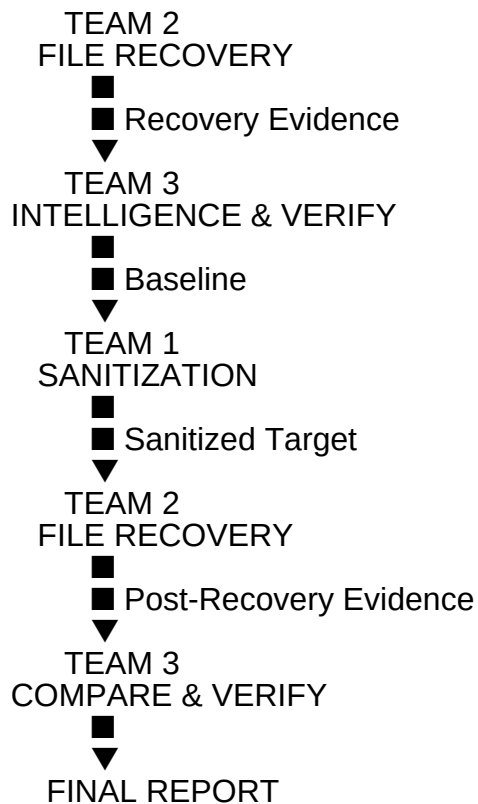
...

---

### # ■ TEAM INTEGRATION

The three teams operate as a closed-loop system.

```text



...

■ Integration Responsibility

| Stage | Responsible Team | Main Output |
|--------------------|------------------|---------------------------------|
| ----- ----- ----- | | |
| Analyze test image | Team 2 | Filesystem/evidence information |
| Find deleted files | Team 2 | Recovered artifacts |


```

■
■■■ README.md
■
■■■ frontend/
■   ■■■ dashboard/
■   ■■■ recovery/
■   ■■■ sanitization/
■   ■■■ verification/
■   ■■■ audit/
■   ■■■ reports/
■
■■■ backend/
■   ■■■ api/
■   ■■■ recovery/
■   ■■■ sanitization/
■   ■■■ verification/
■   ■■■ audit/
■   ■■■ reporting/
■
■■■ recovery/
■   ■■■ sleuthkit/
■   ■■■ carving/
■   ■■■ reconstruction/
■   ■■■ validation/
■   ■■■ signatures/
■
■■■ sanitization/
■   ■■■ file/
■   ■■■ folder/
■   ■■■ disk-image/
■   ■■■ media-classifier/
■
■■■ intelligence/
■   ■■■ confidence/
■   ■■■ scoring/
■   ■■■ baseline/
■   ■■■ comparison/
■
■■■ audit/
■   ■■■ hash-chain/
■
■■■ reports/
■
■■■ certificates/
■
■■■ test-data/
■
■■■ disk-images/
■
■■■ docs/
...

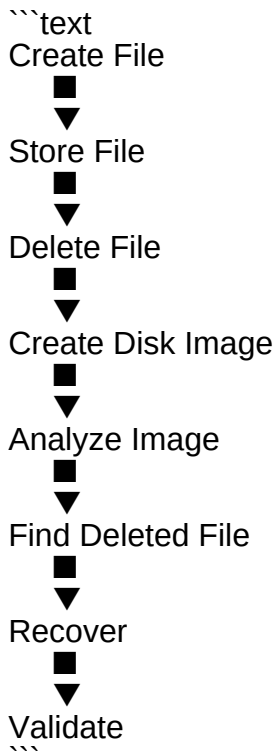
```

> The final repository structure can be modified according to the implementation architecture.

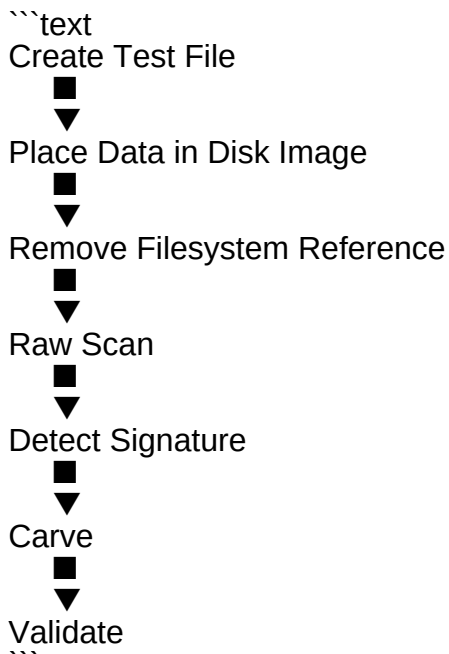
■ TESTING STRATEGY

The MVP should primarily use controlled test disk images.

Test Scenario 1 — Deleted File Recovery



Test Scenario 2 — Raw File Carving



Test Scenario 3 — Sanitization Verification

```

```text
Create Test Image
 █
 ▼
Recover Files
 █
 ▼
Create Baseline
 █
 ▼
Sanitize Image
 █
 ▼
Recover Again
 █
 ▼
Compare Results
 █
 ▼
Verify
 █
 ▼
Generate Certificate
```

```

Test Scenario 4 — Multiple File Formats

The test image can contain:

```

```text
carving.img
 █
 ███ image.jpg
 ███ image.png
 ███ document.pdf
 ███ document.docx
 ███ archive.zip
```

```

The recovery engine attempts to identify and recover each supported artifact.

█ Example Final Result

```

```text
=====
FORENSIC ANALYSIS RESULT
=====

TARGET

Disk Image : carving.img
SHA-256 : <hash>

```

## PRE-SANITIZATION

Files Identified : 5  
Files Recovered : 5  
Files Validated : 5

Recovery Confidence : <score>  
Recoverability : <score>

## SANITIZATION

Method : <method>  
Status : COMPLETED

## POST-SANITIZATION

Files Recovered : 0  
Previously Identified  
Artifacts Recovered : 0

## VERIFICATION

Result : PASS

Statement:  
No previously identified artifacts were recovered  
under the defined verification procedure.

## AUDIT

Audit Chain : VALID

## FINAL STATUS

SANITIZATION : VERIFIED

=====

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## # ■ SECURITY & SAFETY BOUNDARY

The MVP should operate on **\*\*controlled test disk images by default\*\***.

### ## Mandatory Safety Rules

- Never automatically target system disks.
- Never automatically target `/dev/sda`.
- Never automatically target `/dev/nvme\*`.
- Destructive operations require explicit confirmation.
- Display the target path before sanitization.
- Display the SHA-256 hash before sanitization.
- Use controlled test images for destructive demonstrations.
- Do not perform destructive operations on real user data.

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## # ■■ SSD / NVMe Limitation

The prototype should not claim universal or permanent deletion, particularly for:

- SSD
- NVMe
- Flash-based storage

For physical storage types, the MVP should primarily demonstrate:

```
```text
MEDIA CLASSIFICATION
+
METHOD RECOMMENDATION
```
```

rather than automatically wiping real physical devices.

---

## # ■ AI/ML Scope

AI/ML is intentionally **\*\*not included in the working MVP\*\*** due to the short development window.

It can be presented as a future enhancement.

Potential future applications include:

- Intelligent artifact classification
- Advanced recovery prioritization
- Fragmented-file prediction
- Automated evidence classification
- Anomaly detection
- Recovery success prediction

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## # ■ PROJECT UNIQUE SELLING POINTS

### ## 1. Primary USP

> **\*\*Don't just erase data. Prove what was recoverable before sanitization — and prove what remains after**

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### ## 2. Explainable Recovery Confidence

Recovery confidence is supported by understandable forensic evidence such as:

- Valid file signatures
- Structural checks
- Parser validation
- Filesystem evidence

---

### ## 3. Recoverability Score

A project-defined metric that describes how recoverable an artifact is rather than simply reporting that a file

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### ## 4. Before/After Verification

The system establishes a recovery baseline before sanitization and compares it with post-sanitization reco

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### ## 5. Tamper-Evident Audit Trail

Hash-chained audit events make changes to recorded workflow information detectable.

This is an audit-integrity mechanism and **\*\*not blockchain\*\***.

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### ## 6. Automated Reporting & Certificate

The system generates a consolidated result containing:

- Target information
- Sanitization details
- Recovery findings
- Verification outcome
- Audit integrity
- Certificate/report

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## # ■ MVP Scope

The working MVP focuses on:

### ### Team 1

- File sanitization
- Folder sanitization
- Controlled disk-image sanitization
- Storage-media classification
- Sanitization method recommendation
- Safety controls
- SHA-256 capture
- Sanitization reporting

### ### Team 2

- Sleuth Kit integration
- Filesystem analysis
- Deleted-file identification
- Deleted-file recovery
- Raw file carving
- JPEG recovery
- PNG recovery
- PDF recovery
- DOCX recovery
- ZIP recovery
- Fragmented-file reconstruction

- Recovery validation
- Recovery metadata

### ### Team 3

- Explainable recovery confidence
- Recoverability Score
- Pre-sanitization baseline
- Post-sanitization comparison
- Sanitization verification
- PASS/FAIL determination
- Hash-chained audit trail
- Audit integrity verification
- Automated forensic report
- Sanitization certificate

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### # ■ MVP Limitations

The prototype does not claim:

- Universal file recovery
- Guaranteed fragmented-file recovery
- Universal filesystem support
- Permanent deletion on all storage technologies
- Automatic wiping of real physical system drives
- Commercial forensic-suite capabilities
- AI/ML-based recovery during the MVP

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### # ■ Future Enhancements

Possible future improvements include:

- More filesystem support
- More file formats
- Advanced fragmented-file reconstruction
- Advanced file parsers
- Parallel carving
- Large-image optimization
- Duplicate artifact detection
- Advanced evidence visualization
- Advanced recovery scoring
- AI/ML-assisted artifact classification
- Additional storage sanitization standards
- Enterprise-scale reporting
- Cloud-based forensic investigation
- Case management
- Multi-user investigator workflows

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### # ■■ Technology Stack

The prototype may use the following technologies:



## ## Operating System

- Linux
- Ubuntu
- WSL2

## ## Digital Forensics

- The Sleuth Kit
- `mmls`
- `fsstat`
- `fls`
- `istat`

## ## File Recovery

- Raw byte scanning
- File signatures
- Magic bytes
- File structure analysis
- File carving
- File reconstruction
- File validation

## ## Security

- SHA-256
- Hash-chained audit logs
- Integrity verification

## ## Application

- Unified GUI
- Backend/API
- JSON-based metadata
- Automated reporting

---

## # ■ Evidence & Audit Model

The system maintains evidence throughout the entire workflow.

``text

TARGET



■■■ Target Path

■■■ Target Type

■■■ SHA-256



RECOVERY EVIDENCE



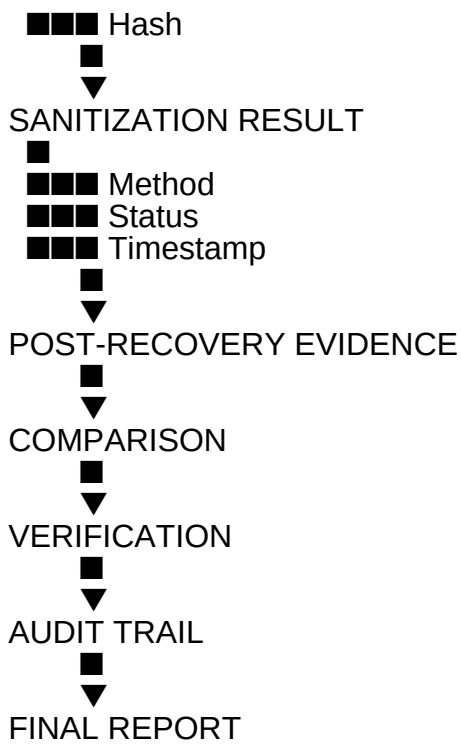
■■■ File Type

■■■ Offset

■■■ Size

■■■ Recovery Method

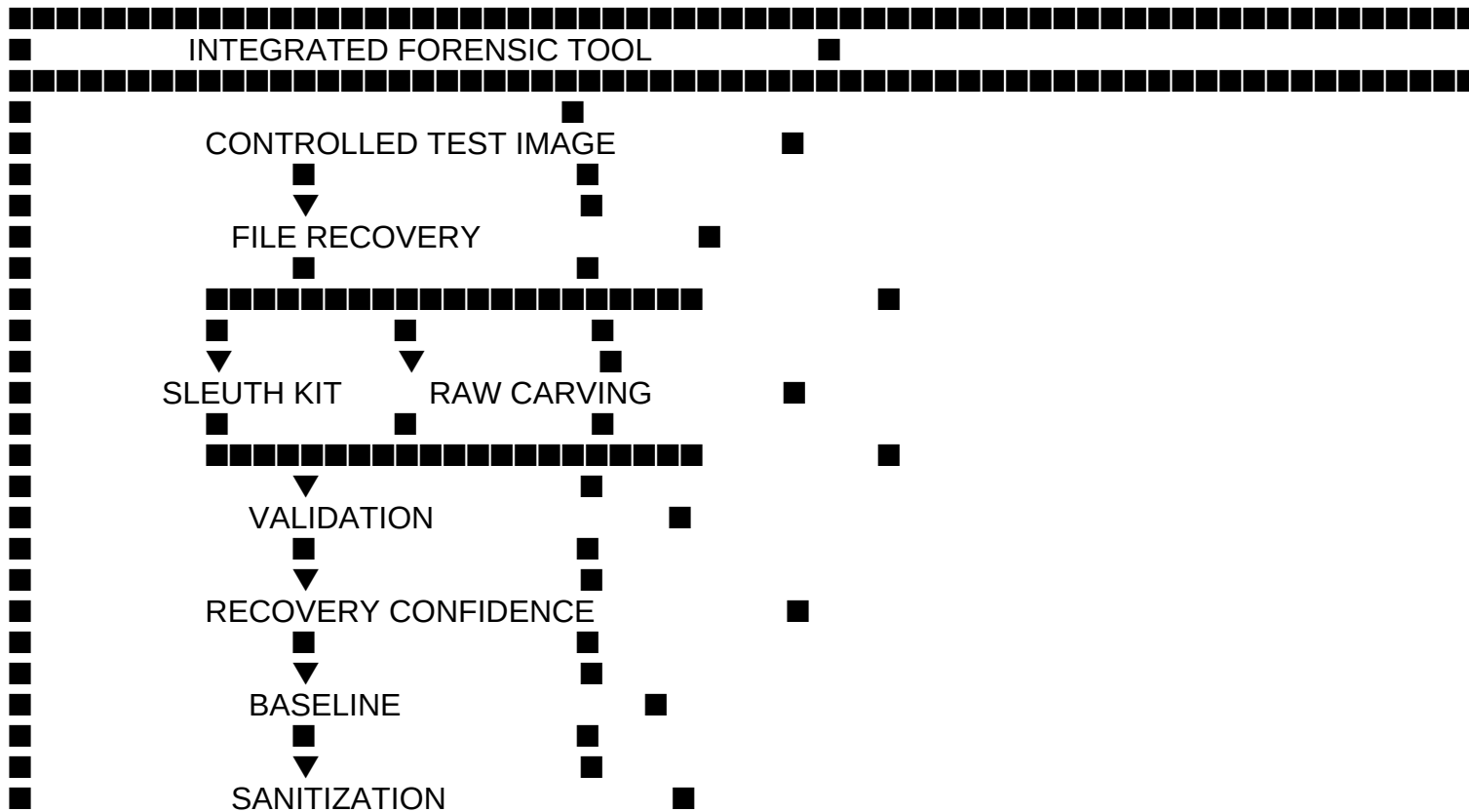
■■■ Validation

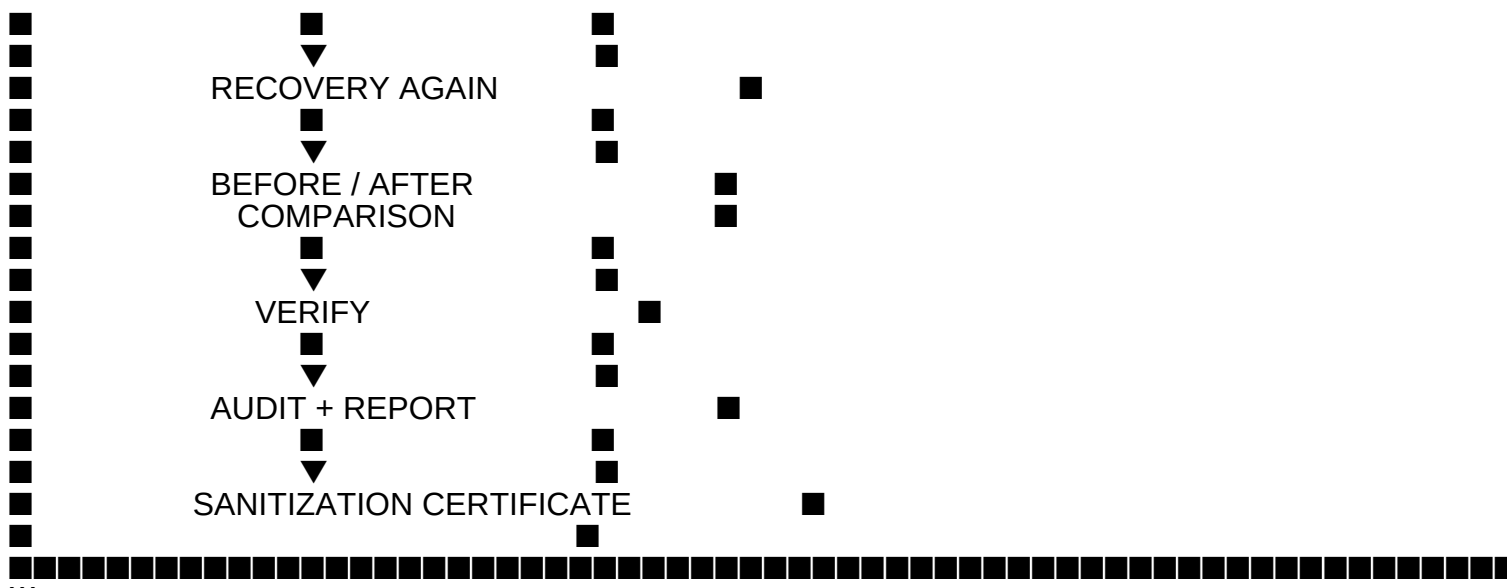


## # Final System Concept

The entire platform can be summarized as:

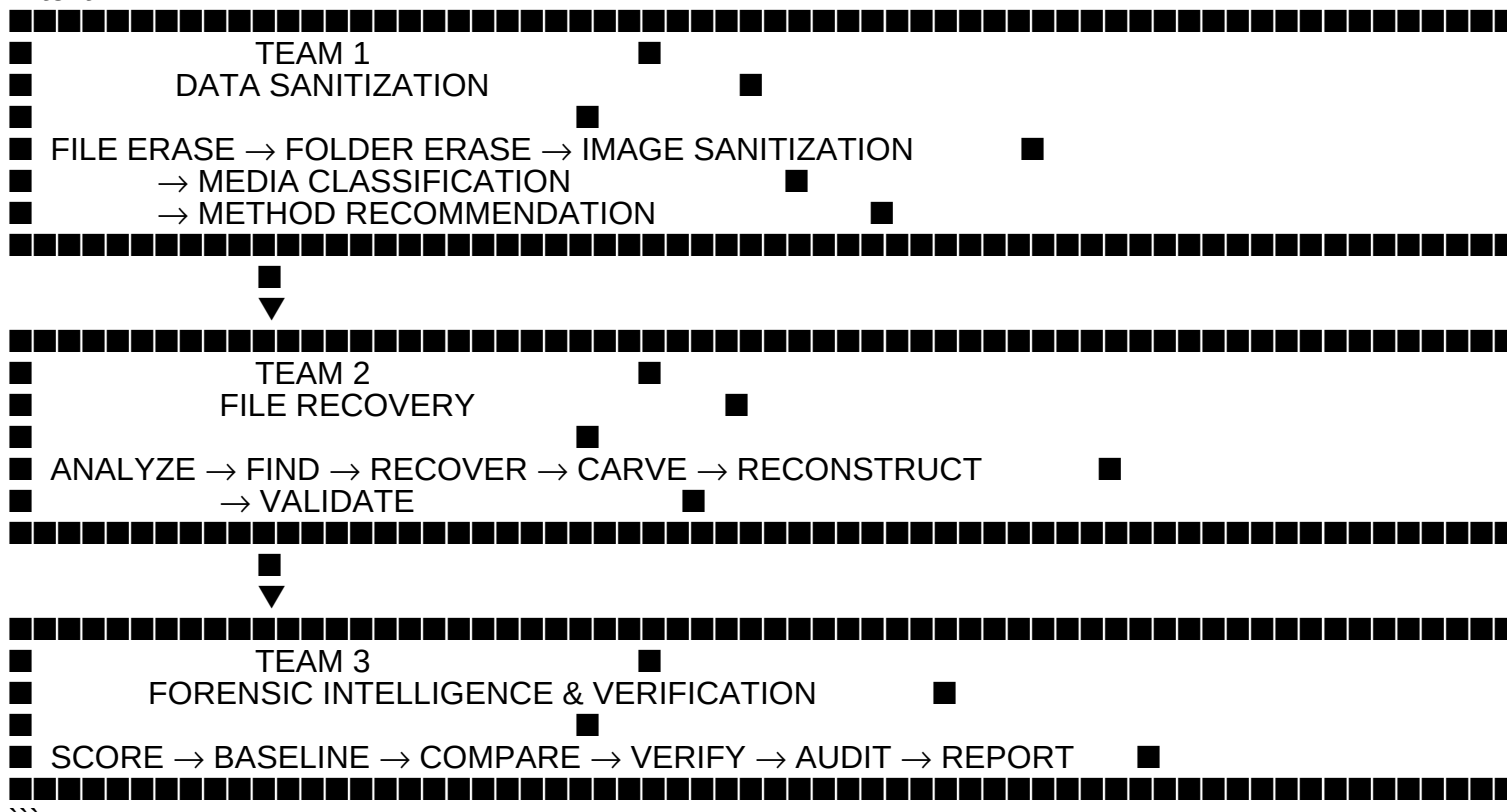
```text





■■■ Team Responsibilities Summary

```text



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## # ■ Final Outcome

The final prototype brings all three teams together into one closed-loop forensic platform.

```text

RECOVER WHAT EXISTS



MEASURE RECOVERY



RECORD BASELINE



SANITIZE DATA



ATTEMPT RECOVERY



COMPARE RESULTS



VERIFY RESULT



PROVE THE OUTCOME



GENERATE REPORT/CERTIFICATE

```

The goal is not merely to erase data or recover files independently.

The goal is to provide an **evidence-driven, measurable, and verifiable workflow** for data sanitization and f

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# ■ License

This project is developed for educational, research, and **Smart India Hackathon (SIH) 2026** prototype p